Automated Student Lab Allocation Using Maximum Bipartite Matching





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DSA 3
Section 7

❋ Abstract



- Laboratory allocation is a common scheduling challenge involving student constraints and limited lab capacities.
- Manual or naïve allocation can result in inefficient assignments, dissatisfaction, and resource underutilization.
- The proposed system models students and laboratories as a bipartite graph, where edges represent student preferences.
- Maximum Bipartite Matching will be used to maximize the number of successful student-lab assignments.
- The system will generate allocation reports and provide visualization of the matching process.
- The project demonstrates the practical application of graph matching algorithms in academic scheduling and resource allocation



Problem Statement :

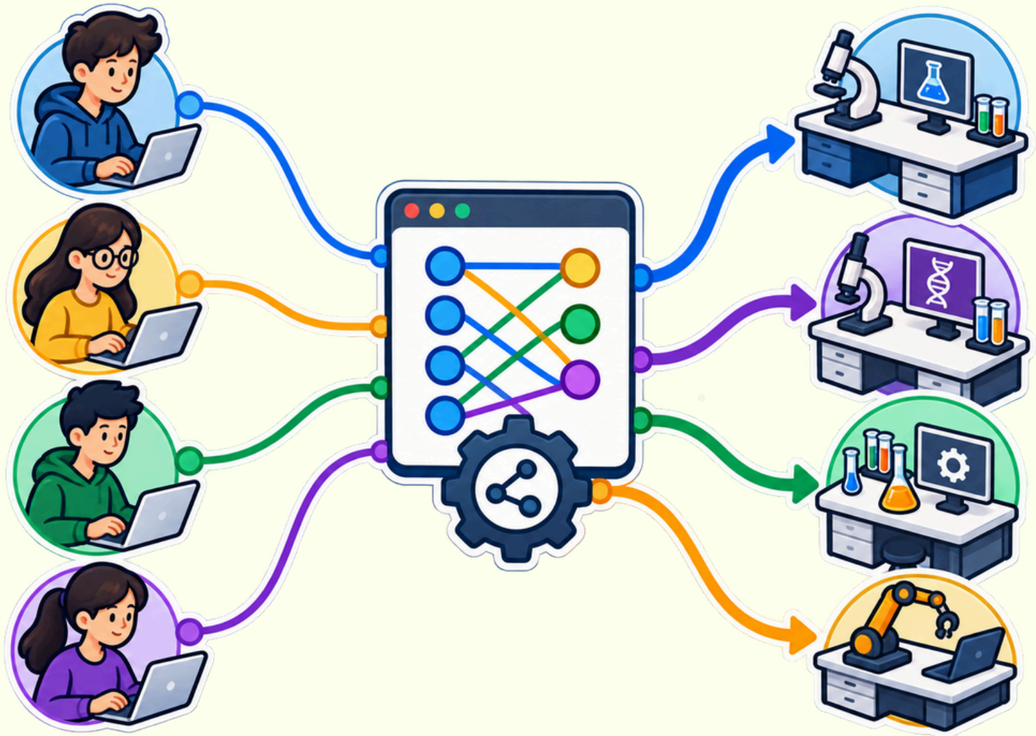
- *Assigning students to laboratory sessions based on capacity constraints is a common scheduling challenge.*
- *Manual or naïve allocation can lead to:*
- *Inefficient assignments*
- *Student dissatisfaction*
- *Underutilization of lab resources*
- *The proposed system will automatically assign students to labs using Maximum Bipartite Matching.*
- *Each student will receive at most one lab allocation.*
- *The system will aim to maximize the total number of successful assignments.*
- *It will also provide allocation reports and visualization of the matching process.*



Proposed * Functionalities *



- Student Management – Enter/Upload student details .
- Lab Management – Add laboratories and define their capacities.
- Bipartite Graph Creation – Represent student–lab preferences as graph edges.
- Maximum Bipartite Matching – Find the maximum number of successful allocations.
- Automatic Allocation – Assign students to suitable labs while respecting constraints.
- Allocation Reports – Display student assignments and lab utilization.
- Graph Visualization – Visualize students, labs, and matched edges.



Tools and Technologies to be used:

Tools :

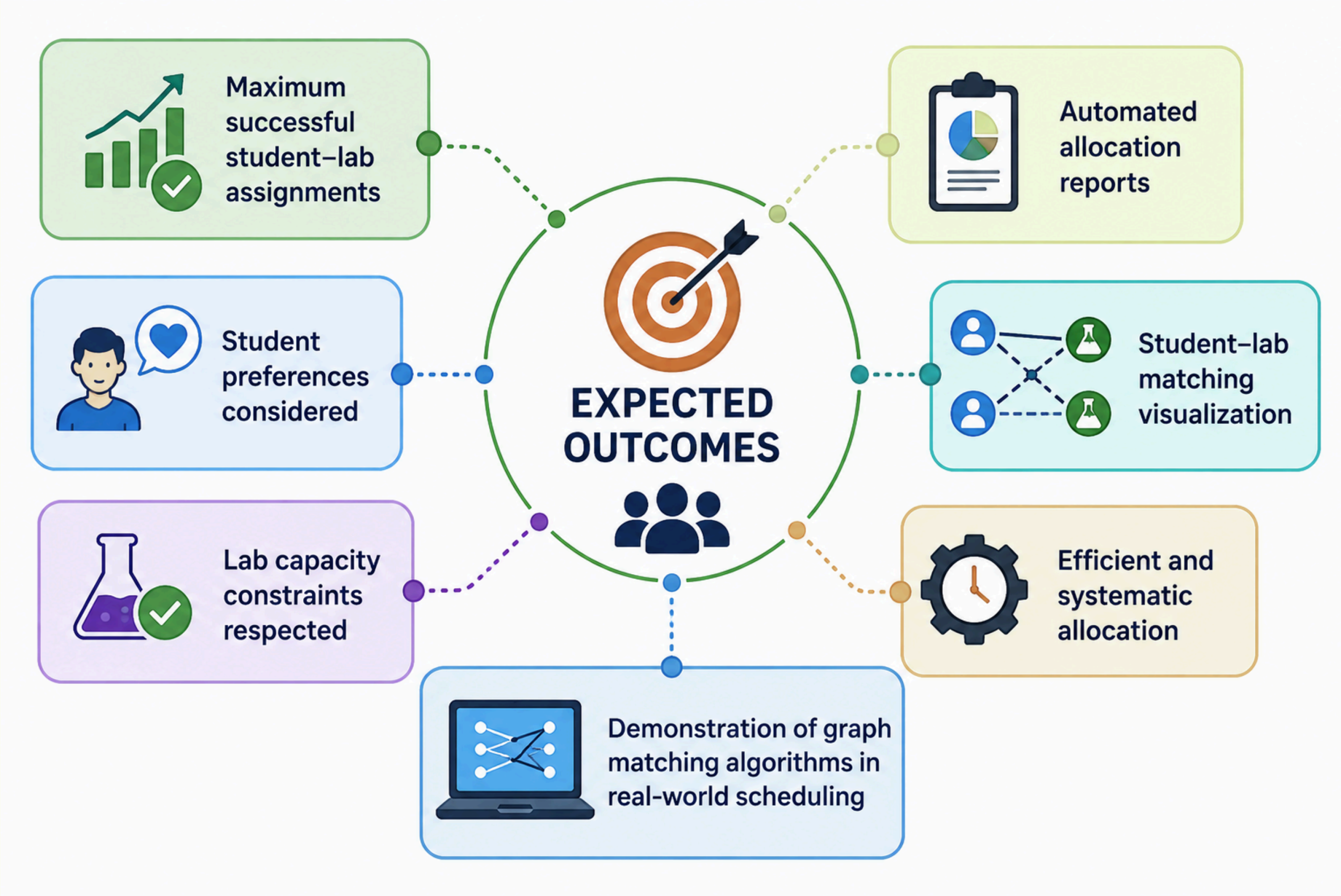
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Java
- **Core Algorithm:** Maximum Bipartite Matching / Hopcroft-Karp
- **Data Structure:** Bipartite Graph
- **Database:** MySQL
- **Graph Visualization:** JavaScript-based visualization
- **Development Environment:** Visual Studio Code / IntelliJ IDEA
- **Version Control:** Git & GitHub

Core Technologies :

Graph Theory + Maximum Bipartite Matching + Database + Web Interface



Expected Outcomes :



Summary :

The Problem

Allocate students to labs with limited seating capacity.


S1


S2


S3


S4


Lab A
Capacity : 2


Lab B
Capacity : 3

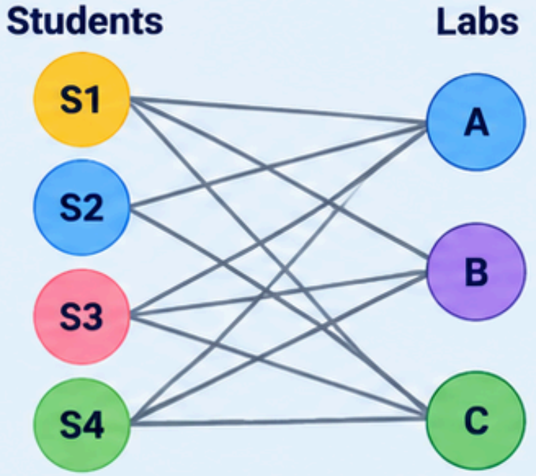

Lab C
Capacity : 2

Manual allocation leads to:

- Inefficient assignments
- Uneven utilization of seats
- Time-consuming process

Our Approach

Maximum Bipartite Matching



Algorithm finds the maximum number of successful assignments while respecting:

- Lab capacities

Result

Student	Assigned Lab
S1	Lab A
S2	Lab C
S3	Lab B
S4	Lab A

Outcome:

- Maximum assignments
- No student in multiple labs
- Lab capacities respected
- Allocation report generated
- Graph visualization provided

Geotag Photo :





Thank You

Team 07